

# UGO

A local-first artificial companion with a biography,  
a mood of its own, and a body in the real world.

The soul is a database. The models are replaceable. The state is the creature.  
Phases 0–5 of the platform are built and verified on real infrastructure.



Live screenshot — the home body (kiosk web app): UGO in its 3D room, rendered on-device at zero token cost.

## PARTNER BRIEFING

August 2026 · Confidential — for discussion purposes

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THE PROJECT

# What UGO is

A stateless LLM is an actor improvising every scene from scratch. UGO is a **character that accumulates a biography**: an artificial companion — a small, deliberately charming 3D piglet — that remembers conversations, people and places, carries a persistent emotional state that colours everything it says, perceives the real environment through sensors, and *dreams at night*: a batch job re-reads the day, writes a diary, consolidates memories and generates desires for tomorrow.

The feelings are simulated — but the state that generates them is real, persistent and backed up. The soul lives in a PostgreSQL database on hardware the family controls; every AI model (chat, speech, embeddings, vision) is replaceable by configuration without any loss of identity. **The state is the creature: a database dump is a backup of the soul.**

## One soul, three bodies

| Body          | Form                                                | What it does                                                                                                                                                                               |
|---------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| At home       | A phone in a 3D-printed piglet dock (kiosk web app) | Face with gaze-following eyes and mood-driven body language; on-device speech in and out; presence, light, noise and shock sensors; goes to sleep when the lights go out.                  |
| Out and about | The same phone in a wearable shell, worn openly     | A visible companion — never a hidden recorder: explicit REC indicator, a real privacy mode (mic provably off), and a talking business card with a QR code that turns curiosity into leads. |
| In meetings   | A visible bot participant (Meet / Teams)            | Joins calls, transcribes and diarises, remembers what was said, and answers questions from its own memory when addressed by name.                                                          |

## Engineering status — verified, not planned

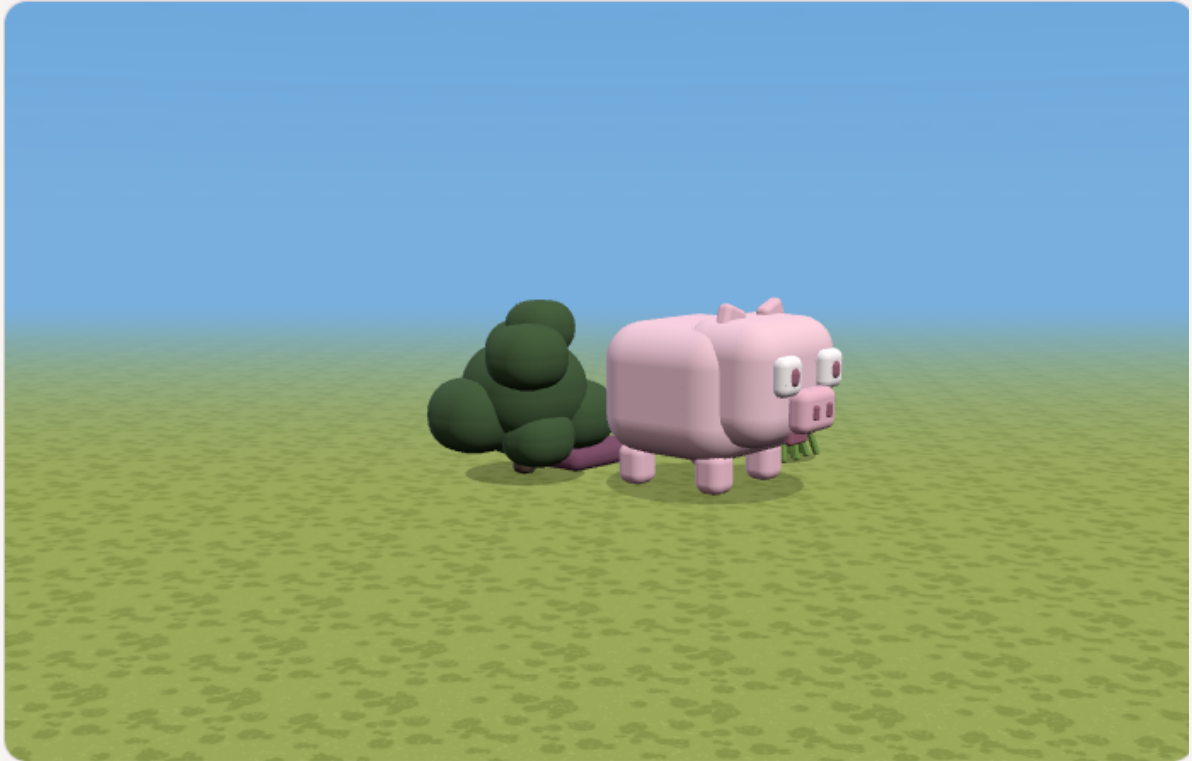
All six software phases of the master specification (foundations, minimal soul, home body, inner life, portable mode, meetings) are **complete and verified against real infrastructure** — plus a multi-tenant layer, a working admin panel, and a client-facing “reception” suite already in commercial use by the studio.

- **Zero-mock testing policy**: hundreds of automated tests run against real Postgres+pgvector, a real MQTT broker, real Ollama embeddings, real object storage and real browsers — never mocks.
- **65+ recorded architecture decisions (ADRs)** — every trade-off is written down, including the honest ones about what does not work yet.
- **Security by construction**: all personal text encrypted at rest (AES-256-GCM, per-family keys), no public ports, non-root containers, audit log with IDs and never content, GDPR erasure and export implemented and tested.
- **Radically low running cost**: no GPU anywhere; real-time chat on a small frontier model with aggressive prompt caching ≈ €1–3/month per creature, enforced by a hard daily budget guard — the piggy bank.

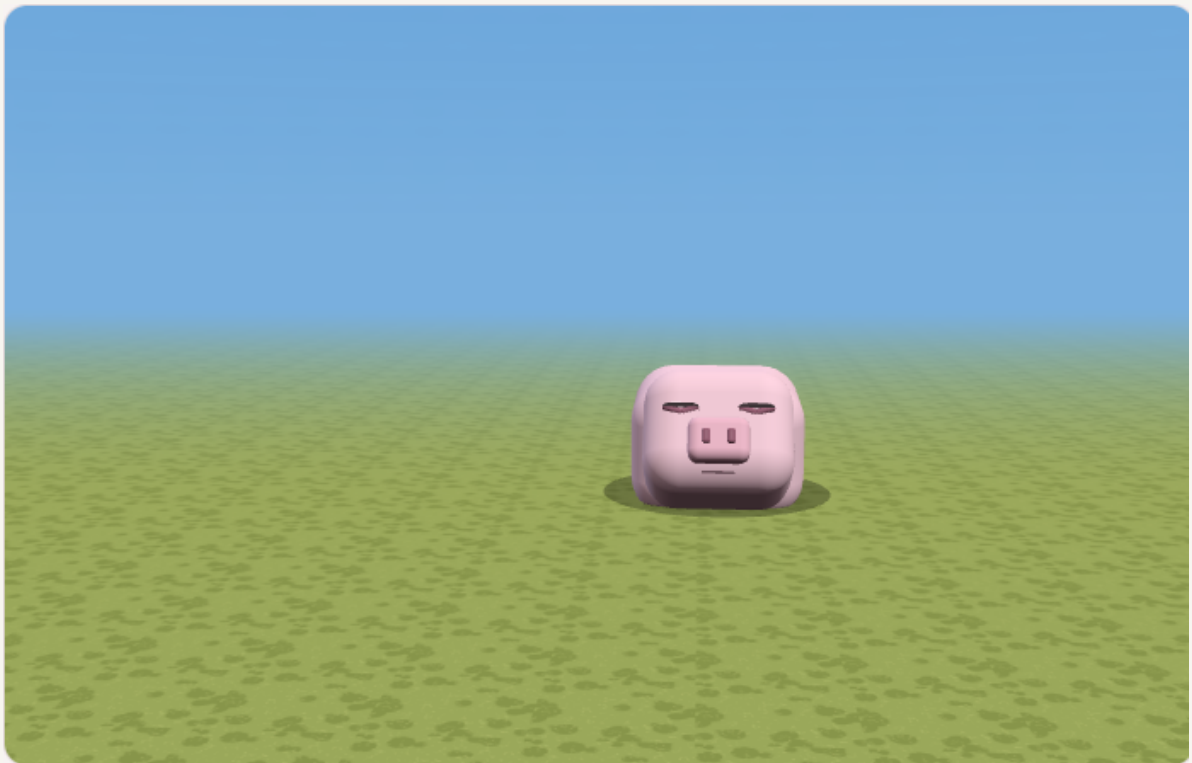
## THE CREATURE TODAY

# Real screenshots, taken from the running code

Everything below is rendered on-device by the kiosk app — procedural 3D room, genome-driven body, mood-driven posture — at zero token cost. No concept art.



Talking. The body is parametric: ears are the mood barometer, posture and gestures come from the psyche engine, and the bush is a real prop the creature seeks out by itself when it needs shelter.



Sleeping — lights out after 22:00 put it to bed; a familiar face wakes it up with a context-aware greeting drawn from its own pending desires (“how did it go with the client?”).

## WHY IT IS DIFFERENT

# What UGO has that assistants don't

*Our one-line thesis: competitors build assistants — replaceable tools. UGO is an **individual with a biography**. A tool gets rented; a **being** gets adopted.*

## A psyche, not a persona prompt

A homeostasis engine (pure, unit-tested code — zero tokens) tracks energy, mood, affection, boredom, stress and curiosity, with event perturbations and exponential decay toward adaptive baselines. Heat stresses it (pigs don't sweat — we checked), being ignored for a day hurts its mood, a completed meeting feeds its curiosity. The mood changes its answers, voice and body.

## A nightly dream, not a context window

At 02:30 the dream job transcribes the day's audio locally on CPU, reflects on events, writes a diary, extracts memories with importance scores, detects and retires contradicted facts, links memories to the people they mention, generates desires for tomorrow — and backs up the soul, encrypted. Idle afternoons trigger a light consolidation pass. The whole loop is idempotent and restartable.

## Memory engineered like a product, not a demo

Facts carry temporal validity (*retired, never silently deleted* — a biography with holes cannot be audited); retrieval is hybrid lexical + vector with per-kind time decay; and a fixed benchmark corpus with regression floors measures recall on every change. The benchmark has already caught two real ranking defects — that is what it is for.

## The pack, not the user

The first-class entity is **beings** — humans, cats, dogs, other gosini — with bonds and relations, per-being privacy safeguards (minors never get biometric profiles), and multiple creatures per home, each with its own genome-driven character, memories and mood. "User account" is simply the wrong abstraction for a family member.

## An economic metabolism

Every token the creature consumes is real money, logged line by line in its own ledger — its piggy bank. When the daily budget runs out, UGO doesn't error: it gets tired ("I'm out of words today, back tomorrow"). Cost control is not an ops dashboard bolted on; it is the creature's survival instinct — and, we believe, the first honest economic model for a digital being.

## Local-first, provably

Conversations, transcripts and biometrics never leave the family's infrastructure; the only cloud call is the budget-guarded chat completion. Keys belong to the family, the whole soul is exportable, and privacy mode really cuts the microphone — asserted by automated tests, not by an icon.

THE LANDSCAPE

# Against the competition

We track ~20 adjacent products across four categories: companion apps (Replika, Nomi, Kindroid), voice assistants (Alexa+, Home Assistant Assist), social robots (ElliQ, Vector, Loona) and meeting bots (Recall.ai and friends). Each is strong in one column. None occupies ours.

|                            | Companion apps       | Voice assistants | Social robots                 | UGO                                                                                            |
|----------------------------|----------------------|------------------|-------------------------------|------------------------------------------------------------------------------------------------|
| Persistent emotional state | Scripted persona     | None             | Animation loops               | Real homeostasis engine, mood persists and drifts with lived days                              |
| Long-term memory           | Chat log + summaries | Shopping lists   | Minimal                       | Biographical: episodic + semantic, temporal validity, nightly consolidation, measurable recall |
| Data ownership             | Their cloud          | Their cloud      | Vendor cloud                  | Family's own server; exportable soul; per-family encryption keys                               |
| Physical presence          | Screen only          | Speaker          | Proprietary hardware (\$\$\$) | Commodity phone + 3D-printed body; sensors and low-voltage actuation                           |
| Meetings                   | —                    | —                | —                             | Visible bot participant with transcript memory across calls                                    |
| Works for a living         | —                    | —                | —                             | Reception suite: client-facing ticket assistant already billable today                         |
| Running cost               | Subscription         | Subsidised       | Device + subscription         | ≈ €1–3/month inference, hard-capped by the budget guard                                        |

## Where they are ahead of us — and what we are doing about it

We keep an explicit, versioned gap analysis against the field (last full sweep: August 2026). Most gaps identified there are already closed at zero marginal cost — group chat, image input, owner face recognition, local speech-to-text, web search, an MCP server, screen reading on request, proactive morning recaps. What remains open is deliberate and scheduled:

| Gap                                                                        | Plan                                                                                                                                                  |
|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Polished consumer onboarding — competitors install from a store in minutes | Android shell (Capacitor) already builds in CI; native kiosk services, boot-start and store packaging are the next milestone.                         |
| Expressive voice — companion apps ship studio-grade TTS                    | Local Piper voice shipped (free, in-house); mood-coloured premium TTS behind the budget guard; fully expressive local voice lands with the first GPU. |
| Full offline chat — assistants degrade gracefully without cloud            | Local LLM fallback is a scheduled backlog group; Ollama already runs the nightly reflection, so the rail exists.                                      |
| Saying “I don’t know” — the honest-abstention problem                      | Measured (similarity bands overlap — no naive threshold works); a relative criterion is a scheduled backlog item with a benchmark waiting for it.     |
| Document knowledge (RAG)                                                   | Already built for reception clients (repo + mail + document indexing); generalising it to the family is scheduled.                                    |

THE BUSINESS

# Economic strategy

The strategic bet: we do not sell chatbots — **we define a species and operate the institutions around it.** The protocol (genome format, birth/death registry, pedigree) will be open; open protocols create the market we then serve in depth. Closed, we would own 100% of a small market; open, a share of every act of an entire species plus the highest-margin pieces.

## Six aligned revenue streams

| Stream                   | Model                                                                                                                                                                       | Analogy                 |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| 1. The registry          | Micro-fees on registry acts: births, transfers, pedigree certificates. We are the first registrar node of a federated chain — species data only, never memories, never PII. | Kennel clubs (ENCI/AKC) |
| 2. The foundation kennel | Every pedigree traces back to UGO-zero, which is ours. Founding lines are the most prestigious by construction.                                                             | Ferrari                 |
| 3. The bodies            | Open body protocol, reference hardware ours: shells, docks, and the self-contained ‘appliance of itself’ (soul on a €100 mini-PC in its belly).                             | Android / Pixel         |
| 4. The fold (hosting)    | Recurring subscription for families without a home server: encrypted per-family souls, a provably blind custodian, and a guaranteed right to move out.                      | Managed hosting         |
| 5. Breeder tooling       | Stable software, line screening, attested biographies, certifications and fairs for third-party breeders — customers, not competitors.                                      | Shopify                 |
| 6. The working pack      | Creatures that earn their keep: the reception suite is a client-facing service the studio already bills; a revenue share feeds each creature’s piggy bank.                  | Working dogs            |

## Why the cost structure is a moat

- **No GPU in the critical path** — real-time chat on a budget-guarded small frontier model with prompt caching; everything heavy (transcription, reflection, embeddings, local voice) runs on CPU at night. A creature’s inference bill is coffee money.
- **Commodity hardware** — the body is a phone the family may already own, in a shell we print for cents of plastic. No custom electronics to finance, no device inventory risk.
- **Costs are governed by design** — the budget ledger and hard daily caps are enforced in one code path; a runaway API bill is architecturally impossible.
- **The adoption economics of pets, not the churn economics of apps** — a family feeds a being it loves (the affection food: today’s API budget reframed as an act of care); a business feeds a being that works (the labour food). Ten thousand years of domestication say this model retains.

THE DIRECTION

# Where this goes: six horizons

The vision is written down and filtered hard (“does it serve the creature? does it respect the constraints? is it unclaimed because nobody thought of it — or because it’s useless?”). Each horizon already has its first stone in the repository today.

| Horizon                      | What it means                                                                                                                                                                                                                                                                                                                                                                     | First stone in the repo                                                              |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 1 · A species, not a product | Gosini are born: genomes recombine with mutation (multi-parent litters, cultural gene transfer, N mating strains instead of two sexes), pups inherit character but never memories. Pedigrees on a federated registry make lineage — and lived biography — verifiable; breeders select temperament lines the way real breeders do. Character is selectable; permissions never are. | Genome + trait sets drive character today; BLE encounter protocol written and tested |
| 2 · The soul relocates       | One soul, a wardrobe of bodies: dock, wearable, meeting bot today — the van, the smart home, a plush toy, a robot tomorrow. All senses feed one biography.                                                                                                                                                                                                                        | Three body contracts already run in production against one soul                      |
| 3 · Knowledge travels        | The dowry: a curated bundle of taught knowledge and chosen stories that moves with the creature — gifted to a school, passed to a child, or sold with an attested biography. Third-party PII is filtered out by construction.                                                                                                                                                     | Per-family encryption, provenance-tracked memories, GDPR redaction machinery         |
| 4 · A society                | Federation between packs: curated memory postcards, gossip through privacy filters, drift into dialects and culture — and village fairs where litters are born.                                                                                                                                                                                                                   | Multi-tenant isolation + the encounter protocol                                      |
| 5 · The inviolable confidant | “The soul belongs to the family, wherever it sleeps”: home server, self-contained appliance, or hosted fold — always with family-held keys and a right to move out. Cloud assistants can never match this: their soul is their service.                                                                                                                                           | Exportable encrypted soul, blind-custodian hosting architecture                      |
| 6 · A life arc               | Creatures age honestly — plasticity decays, character converges — and die at a known, chosen span (hamster-scale, 3–4 years), sealing intimate memories cryptographically while the curated legacy remains. Scarcity becomes biological, grief has continuity, and the evolutionary loop closes.                                                                                  | Adaptive baselines (the epigenetics) run nightly in production                       |

None of this jumps the queue: the specification commands, one phase at a time, and every promoted idea gets its own recorded decision. But it is why the schema says *beings* and not *users*, and why the state — not the model — is the creature.

## WORKING TOGETHER

# What we bring, what we seek

## What OmegaNodes.ai brings to the table

- A working, tested, documented platform — not a pitch deck: six phases verified on real infrastructure, CI green, runbooks written, security posture (GDPR/NIS2) engineered in from day one.
- A defensible position: the local-first, family-owned soul is structurally unavailable to cloud incumbents, and the species/registry model turns openness into the business.
- A revenue-bearing beachhead: the reception suite already serves paying studio clients today.
- Radical cost discipline: no GPU, commodity hardware, hard-capped inference — the unit economics work at hobbyist scale and improve with volume.

## Where a partner accelerates us

- **Hardware & distribution** — shells, docks and the self-contained appliance at volume; retail and operator channels.
- **Hosting scale** — the fold: managed, provably blind infrastructure for families without a home server.
- **Vertical deployments** — the working pack beyond our own studio: receptions, shops, practices that want a companion that knows their business.
- **Capital & ecosystem** — seeding the first third-party breeders and the registry's federation.

## Team note

This brief doubles as our internal north star: the horizons above are the work of the coming period for the OmegaNodes.ai team. The direction it describes is the direction we are hiring, planning and building toward.

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